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Alliance School of Advanced Computing Bengaluru

Department of CSE

5CS1990

E1CSA313

Design Project-I Diary

Project/Batch ID	DP1-24-28-CSE-GEN-J-07		
Name of the Mentor	Dr. Jayabhaduri Radhakrishnan		
Title of the Project	AI-Driven Smart Traffic Violation Detection and Alert System		
Domain (Area)	Artificial Intelligence & Computer Vision		
Name of the Leader	Sanjay N		
Details of the students	**Name**	**Registration Number**	**Branch**
	Vivek H	2411021061234	B. Tech CSE
	Mohammed Aatif Zafar	2411021061282	B. Tech CSE
	Sanjay N	2411021061200	B. Tech CSE
	Pavan urs B K	2411021061272	B. Tech CSE

Sign. of the Mentor **Sign. of the Project Coordinator-Dept.** **Signature of the Chief Coordinator-ASAC**

Project Details

Title of the Project: AI-Driven Smart Traffic Violation Detection and Alert System

Description: An AI-based computer-vision system that analyses video from junction CCTV cameras and dashcams on Bengaluru roads to detect traffic violations automatically. A YOLOv8 deep-learning model, fine-tuned on Indian traffic data, detects vehicles, riders, helmets, seatbelts, mobile phones, number plates and the traffic-signal state in each frame. ByteTrack assigns a unique ID to every road user so each violation is counted once, and a rule engine decides whether a violation has occurred using virtual stop-lines, signal state, lane direction and rider count. The number plate of the offending vehicle is read using OCR with multi-frame voting, and an evidence-backed alert is sent to the offender and the traffic police through SMS/app notifications and a live web dashboard.

Scope: The project covers automatic detection of six common violations: riding without a helmet (rider and pillion), triple riding, red-light jumping, wrong-way driving, driving without a seatbelt and mobile-phone use while driving. For every violation, the system produces a record with an evidence clip, number plate, time and location, sends an SMS/app alert, and shows the event on a police dashboard with hotspot maps. It runs on edge hardware (NVIDIA Jetson Orin Nano or a laptop GPU), sending only violation events to the server. Integration with the official e-challan payment system and legal processing is outside the scope of this project.

Objectives:

1. To study existing AI-based traffic violation detection systems and identify their limitations.
2. To build a single deep-learning pipeline that detects six common traffic violations from Bengaluru road video.
3. To achieve at least 85% mAP@0.5 for helmet, rider and number-plate detection on Indian test data, at 15+ FPS on edge hardware.
4. To read Indian number plates, including two-line and HSRP plates, accurately using multi-frame OCR voting.
5. To send an evidence-backed alert within a few seconds of a violation and log it on a web dashboard.
6. To analyse violation hotspots and peak hours to help the police plan enforcement.

Outcomes:

- A working prototype that detects six traffic violations from junction and dashcam video in near real time.
- Automatic vehicle identification through Indian number-plate recognition with multi-frame OCR voting.
- Evidence-backed violation records (clip, number plate, time and location) with instant SMS/app alerts to offenders and the traffic police.
- A web dashboard with a live violation feed, search by number plate, hotspot heat maps and peak-hour charts.
- Reduced manual monitoring and more efficient traffic surveillance, validated using mAP, precision, recall, F1-score and FPS.

Problem Statement: AI-Based Traffic Violation Detection and Alert System Using Computer Vision

Bengaluru's roads carry dense, mixed traffic of two-wheelers, cars, autorickshaws and buses, and the city records lakhs of traffic violations every month. Between January and April 2025, riding without a helmet (9.8 lakh cases), not wearing a seatbelt (2.5 lakh) and signal jumping (2 lakh) were the most common offences. The Bengaluru Traffic Police use an Intelligent Traffic Management System (ITMS) with 250 ANPR cameras and 80 red-light violation cameras, but these cover only 50 junctions, leaving many roads unmonitored. Manual policing cannot watch every junction, and offenders are informed only after the fact.

Existing research also has clear gaps. Most systems handle only one or two violation types, and many stop at detection without reading the number plate or sending alerts. Recall remains modest (about 60–73%) in dense and occluded traffic, plate readers trained on foreign plates struggle with Indian two-line and non-standard plates, and night-time, rain and edge-device conditions are rarely evaluated.

Therefore, there is a need for an affordable, extensible AI system that detects multiple violation types from video in real time, identifies the vehicle from its number plate, and alerts both the offender and the traffic police with evidence. Such a system should work with both fixed junction cameras and vehicle dashcams, run on low-cost edge hardware, and turn violation data into hotspot and peak-hour insights that help the traffic police plan where and when to deploy personnel.

Abstract:

Traffic violations such as riding without a helmet, triple riding, signal jumping, wrong-way driving, not wearing a seatbelt and using a mobile phone while driving are common on Bengaluru's roads, and fixed junction cameras and manual checks cannot monitor them everywhere. This project proposes an AI-Driven Smart Traffic Violation Detection and Alert System that turns traffic video into verified violation alerts. Video from junction CCTV cameras and vehicle dashcams is pre-processed and passed to a YOLOv8 detector fine-tuned on Indian datasets (IDD, AI City 2023 Track 5 and RideSafe-400) and on clips collected from Bengaluru junctions. ByteTrack assigns an ID to each road user and links riders to their motorcycles, and a rule engine checks each track against stop-lines, signal state, lane direction and rider count. The number plate of the offending vehicle is read using OCR with Indian plate-format checking and majority voting across frames. For each violation, an evidence clip and record are stored, an SMS/app alert is sent to the offender, and the traffic police view the event on a live dashboard with hotspot maps. The system is designed to run on edge hardware such as the NVIDIA Jetson Orin Nano, sending only violation events to the server to save bandwidth and protect privacy. Performance will be evaluated using mAP, precision, recall, F1-score, plate-recognition accuracy, frames per second and alert latency. Overall, the project aims to reduce manual monitoring and help the traffic police plan enforcement more effectively.

Weekly Meeting Record

Week No: _____

Date: _____ to _____

Design Project Title: AI-Driven Smart Traffic Violation Detection and Alert System

Group Members:

1. Name: Sanjay N Reg. No. 2411021061200 ☐ Present
2. Name: Vivek H Reg. No. 2411021061234 ☐ Present
3. Name: Mohammed Aatif Zafar Reg. No. 2411021061282 ☐ Present
4. Name: Pavan urs B K Reg. No. 2411021061272 ☐ Present

Meeting Venue: LT-2, OF-202 ☐ On Time

Documents Submitted:

☐ Status Report

☐ On Time

Progress:

Issues:

Assessment of Progress:

☐ Excellent

☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the Mentor

Weekly Meeting Record

Week No: _____

Date: _____ to _____

Design Project Title: AI-Driven Smart Traffic Violation Detection and Alert System

Group Members:

1. Name: Sanjay N Reg. No. 2411021061200 ☐ Present
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Notes / Concerns / Comments: _____

Signature of the Mentor

Weekly Meeting Record

Week No: _____

Date: _____ to _____

Design Project Title: AI-Driven Smart Traffic Violation Detection and Alert System

Group Members:

1. Name: Sanjay N Reg. No. 2411021061200 ☐ Present
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3. Name: Mohammed Aatif Zafar Reg. No. 2411021061282 ☐ Present
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Meeting Venue: LT-2, OF-202 ☐ On Time

Documents Submitted:

☐ Status Report

☐ On Time

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Issues:

Assessment of Progress:

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☐ Good

☐ Satisfactory

☐ Fair

☐ Poor

Notes / Concerns / Comments: _____

Signature of the Mentor

Review-I Evaluation Form

Date of Review:

Reviewer Comments: (*Provide specific feedback on Problem identification, Need and scope, Literature review and gap analysis, Objectives and requirements, Proposed methodology, tools and feasibility and Presentation and technical defence)*

Recommendations for Improvement: (*List suggestions or required changes before the next phase/review.)*

Mentor Signature	Examiner 1 Signature	Examiner 2 Signature
Name:	Name:	Name:
Date:	Date:	Date:

Review-II Evaluation Form

Date of Review:

Reviewer Comments: (*Provide specific feedback on the System architecture, Module description, Implementation progress — target 50% of the approved scope, Presentation and viva and Project demonstration)*

Recommendations for Improvement: (*List suggestions or required changes before the next phase/review.)*

Mentor Signature	Examiner 1 Signature	Examiner 2 Signature
Name:	Name:	Name:
Date:	Date:	Date:

Review-III Evaluation Form

Date of Review:

Reviewer Comments: (*Provide specific feedback on the Implementation completeness, Demonstration, Usability and user satisfaction, Submission of the paper and Report quality and conformance)*

Recommendations for Improvement: (*List suggestions or required changes before the Final Presentation in Semester End Examination.)*

Mentor Signature	Examiner 1 Signature	Examiner 2 Signature
Name:	Name:	Name:
Date:	Date:	Date:

Semester End Examination Evaluation

Date of Review:

Reviewer Comments: (*Provide specific feedback on the Project Report submitted, Problem Formulation, Review of Literature, Design Methodology, and Concept of the Work / Originality, Significance & Relevance - Application in the Real World and Implementation in terms of Presentation, Demonstration and Viva Voce)*

Project Submission Checklist

Item	Yes	No	Remarks
Project Report Submitted	☐	☐	
Survey Paper Submitted in Conference	☐	☐	
Source Code / Software Submitted (if applicable)	☐	☐	
Prototype or Model Demonstrated (if applicable)	☐	☐	
Presentation Slides Submitted	☐	☐	
Project Diary Maintained	☐	☐	
Ethical Clearance Obtained (if applicable)	☐	☐	
Any Other Supporting Documents	☐	☐	

Examiner 1 Signature

Name:

Date:

Examiner 2 Signature

Name:

Date:

Student Consent Form

Academic Year: 2026-27

Semester: V

Degree Program: B.Tech in CSE

Project Title: AI-Driven Smart Traffic Violation Detection and Alert System

Mentor Name: Dr. Jayabhaduri Radhakrishnan

Mentor's Designation: Professor

Team Members

Sl. No.	Name of the Student	Register Number
1.	Vivek H	2411021061234
2.	Mohammed Aatif Zafar	2411021061282
3.	Sanjay N	2411021061200
4.	Pavan urs B K	2411021061272

Declaration by the Students

We hereby confirm that we have chosen the above topic for our Design Project – I as part of the curriculum for the **V** semester of our **B.Tech** degree program.

We understand and agree to the following:

1. The project work is academic in nature and must follow the guidelines.
2. We will carry out the project work ethically and responsibly under the supervision of the assigned Mentor.
3. We will not engage in plagiarism or submit any copied or previously submitted work.
4. We will attend all review meetings and submit documentation as per the department schedule.
5. Any form of academic misconduct or non-compliance may lead to disqualification or disciplinary action.

Signatures of Students

Sl. No.	Name of the Student	Signature
1.	Vivek H	
2.	Mohammed Aatif Zafar	
3.	Sanjay N	
4.	Pavan urs B K	

Design Project – General Rules

1. All projects must be approved by the supervising faculty before starting.
2. All work must be original; plagiarism is strictly prohibited.
3. Students can work in teams.
4. Teams must have 4 members.
5. Roles and contributions of each team member must be clearly defined.
6. A team leader must be designated for communication and coordination.
7. Team members must contribute fairly and equally.
8. Conflicts must be reported promptly to the Faculty In-charge / Coordinator.
9. Students must meet regularly with their Mentor and Faculty In-charge of the class weekly.
10. Meeting notes or progress reports must be maintained.
11. Follow official timelines for proposals, drafts, final reports, and presentations.
12. Maintain the documentation phase by phase.
13. Projects must follow ethical guidelines (no falsified data, respect IP, consent/privacy for surveys).
14. Follow the formatting guidelines (APA, IEEE, etc.) for all submissions.
15. Submit all required documents: proposal, report, logbook, and presentation materials.
16. Deliver a final presentation and be prepared for questions.
17. Maintain professional conduct with Mentor, Faculty In-charges, Teammates, and evaluators.
18. Use institutional resources responsibly and uphold academic integrity.